

Revision · Symmetry in three dimensions

Cambridge insert: planes of symmetry, axes of rotation, and the shapes that have them.

LESSON 67

1 × 40 MIN

GEOMETRY 8 – REVISION

STAGE 9 · 14.3 + PROJECT 5

LESSON CLOCK

5'

14'

12'

5'

4'

Find the mirror planes of the classroom	5 min
Planes of symmetry	14 min
Axes of rotation	12 min
Project 5	5 min
Homework	4 min

LEARNING OBJECTIVES

- Identify planes of symmetry in a solid.
- Identify axes of rotational symmetry and their order.
- Compare the symmetry of familiar solids.
- Use symmetry to solve a design problem (Project 5).

TEXTBOOK

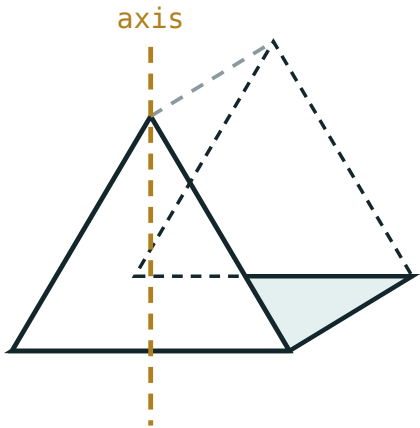
National	Annual revision
Cambridge	Learner’s Book pp. 335–342
Workbook	Workbook 14.3

01

Explanation

Two kinds of symmetry in space

In the plane a shape has **lines** of symmetry. In space it has **planes** of symmetry: a flat cut that leaves each half a mirror image of the other.



a plane of symmetry cuts it into mirror halves

A cuboid has three planes of symmetry, each cutting it exactly in half.

It may also have **axes of rotational symmetry**: a line about which the solid can be turned and still look the same. The **order** counts how many positions look identical in one full turn.

SOLID	PLANES OF SYMMETRY	ROTATION AXES
cuboid (all edges different)	3	3, each of order 2
square-based prism	5	1 of order 4, 4 of order 2
cube	9	3 of order 4, 4 of order 3, 6 of order 2
cylinder	infinitely many	1 of infinite order
sphere	infinitely many	infinitely many

The pattern is clear: the more regular a solid, the more symmetry it has. The sphere, the most regular of all, is symmetric about every plane through its centre.

Project 5 · Designing with symmetry

Task. Design a container of capacity *1 litre* that has at least two planes of symmetry, and calculate the area of material it needs.

1. Choose a shape — a cuboid, a square prism, a cylinder, a hexagonal prism.
2. Fix the dimensions so that the volume is 1000 cm^3 .
3. Calculate the surface area.
4. Count and describe every plane of symmetry.
5. Compare with a neighbour: whose design uses least material?

THE RESULT WORTH DISCOVERING

Among all cuboids of a given volume, the **cube** uses least material; among all cylinders, the one with $h = 2r$. And of all shapes whatever, the sphere is the most economical — which is why bubbles are round.

02 Worked examples

EXAMPLE 1 How many planes of symmetry has a cuboid with all three edges different?

- 1 One parallel to each pair of opposite faces.
Each cuts the solid exactly in half.

- 2 Three pairs of faces.

- 3 3 planes.
A diagonal cut does **not** work unless two edges are equal.

ANSWER

3

EXAMPLE 2 A cylinder of capacity 1 litre has $h = 2r$. Find r and its surface area.

- 1 $\pi r^2 \cdot 2r = 1000$
 $1 \text{ l} = 1000 \text{ cm}^3$.

- 2 $2\pi r^3 = 1000, r^3 \approx 159.2$

- 3 $r \approx 5.42 \text{ cm}, h \approx 10.84 \text{ cm}$

- 4 $S = 2\pi r^2 + 2\pi rh \approx 184.5 + 369.1$

- 5 $\approx 554 \text{ cm}^2$

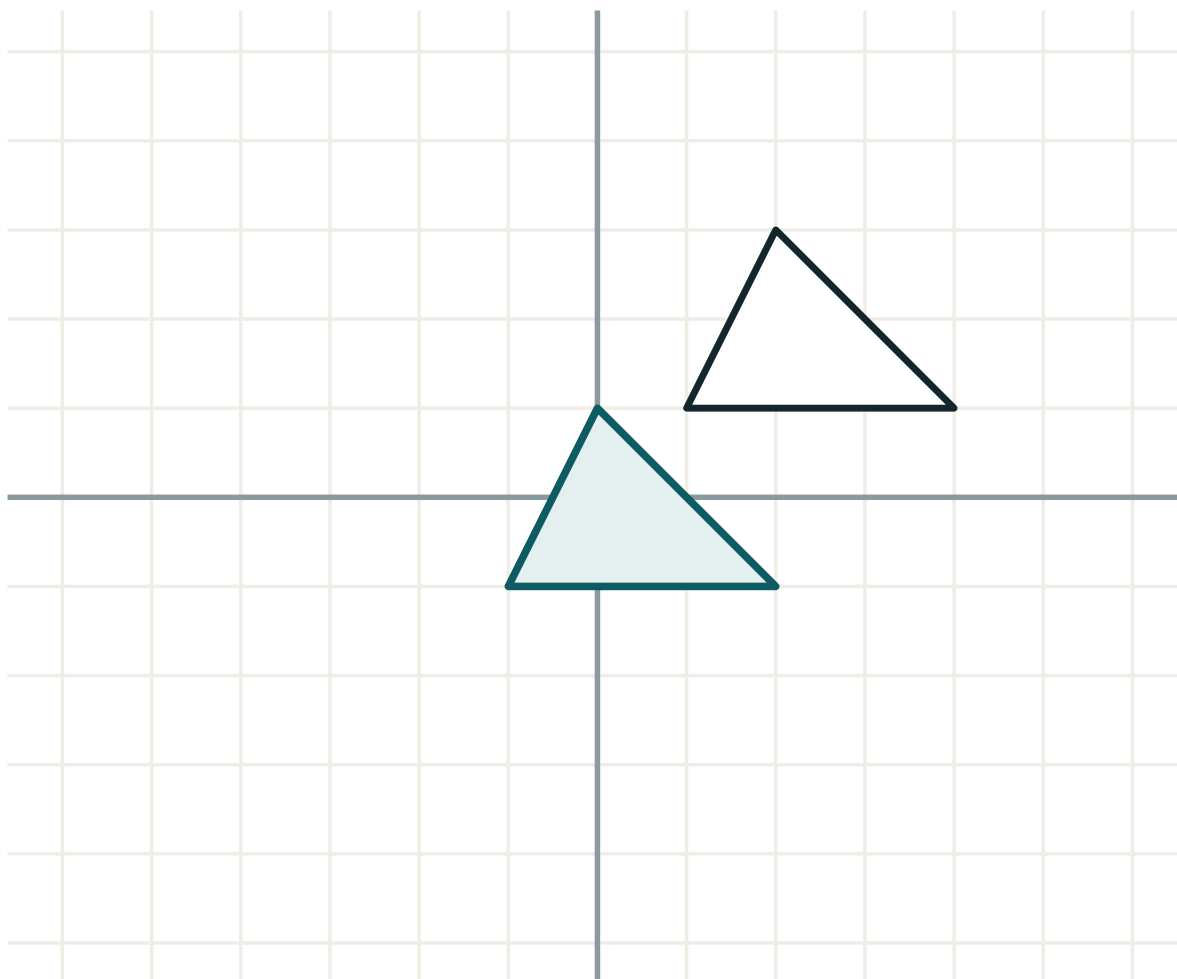
ANSWER

$r \approx 5.4 \text{ cm}, S \approx 554 \text{ cm}^2$

03 Interactive model

Use the transformation model to check which reflections leave a shape looking unchanged.

Reflections that leave a shape unchanged



transformation **translation**

lengths **unchanged**

angles **unchanged**

orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Plane of symmetry	Simmetriya tekisligi	Плоскость симметрии
Axis of symmetry	Simmetriya o'qi	Ось симметрии
Rotational symmetry	Burilish simmetriyasi	Поворотная симметрия
Order of symmetry	Simmetriya tartibi	Порядок симметрии
Solid (3-D shape)	Jism	Тело (пространственная фигура)
Cross-section	Kesim	Сечение
Sphere	Sfera	Сфера
Regular solid	Muntazam jism	Правильное тело

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A cuboid with all edges different has how many planes of symmetry?

A cube has how many planes of symmetry?

A sphere has:

A cylinder has an axis of rotational symmetry of order:

2

4

infinite

none

Of all cuboids of a given volume, least material is used by:

a long thin one

a flat one

the cube

they are all equal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *How many planes of symmetry has a cube?*

ANSWER 9

2. *How many planes of symmetry has a sphere?*

ANSWER Infinitely many.

3. *How many planes of symmetry has a cylinder?*

ANSWER Infinitely many — every plane through its axis, plus the one across the middle.

4. *Name a solid with exactly 3 planes of symmetry.*

ANSWER A cuboid with all edges different.

5. *What is the order of rotational symmetry of a square-based prism about its long axis?*

ANSWER 4

6. *Has a cuboid an axis of rotational symmetry?*

ANSWER Yes — three, each of order 2.

7. *Find the volume of a cube of side 10 cm in litres.*

ANSWER 1 litre

Medium · the standard question

7 PROBLEMS

1. *A square-based prism has how many planes of symmetry?*

ANSWER 5

2. *A cuboid $4 \times 4 \times 6$ — how many planes of symmetry?*

ANSWER 5

3. *Design a cuboid of volume 1000 cm^3 with 5 planes of symmetry, and find its surface area.*

ANSWER $10 \times 10 \times 10$ has 9 planes; $5 \times 5 \times 40$ has 5, with $S = 850 \text{ cm}^2$

4. *A cube of side 10 cm — find its surface area.*

ANSWER 600 cm^2

5. *A cylinder of radius 5 and height 12.7 has volume $\approx 1000 \text{ cm}^3$. Find its surface area.*

ANSWER $\approx 157 + 399 = 556 \text{ cm}^2$

6. *Which uses less material for 1 litre: a 10 cm cube or a $5 \times 5 \times 40$ cuboid?*

ANSWER The cube — 600 cm^2 against 850 cm^2 .

7. *What is the order of rotational symmetry of a regular hexagonal prism about its axis?*

ANSWER 6

Hard · stretch and reason

7 PROBLEMS

1. *A cuboid has volume 1000 cm^3 and a square base. Find the dimensions using least material.*

ANSWER The cube $10 \times 10 \times 10$, $S = 600 \text{ cm}^2$

2. *Show that a cylinder of volume V uses least material when $h = 2r$.*

ANSWER Testing $r = 4, 5.42, 7$ for $V = 1000$ gives $S \approx 601, 554, 594$ — the minimum is at $h = 2r$.

3. *A cube of side 6 is cut in half by a plane through two opposite edges. Is the cut a plane of symmetry?*

ANSWER Yes — it is one of the six diagonal planes of the cube.

4. *Compare the surface area of a sphere of radius 6.2 cm ($\approx 1000 \text{ cm}^3$) with the best cylinder and the best cube.*

ANSWER Sphere $\approx 483 \text{ cm}^2$, cylinder ≈ 554 , cube 600 — the sphere wins.

5. *How many axes of rotational symmetry has a cube, and of what orders?*

ANSWER 13 — 3 of order 4 (face to face), 4 of order 3 (corner to corner), 6 of order 2 (edge to edge).

6. *A regular hexagonal prism — how many planes of symmetry?*

ANSWER 7 — six through the axis, one across the middle.

7. *Explain why bubbles are spherical.*

ANSWER Surface tension pulls the film to the smallest possible surface area for the air enclosed, and of all shapes of a given volume the sphere has the least surface area.

07 Homework

Homework — Project 5

Cambridge Stage 9, Unit 14.3 and Project 5. Bring your design to the next lesson.

1. *Design a container of capacity exactly 1 litre with at least two planes of symmetry.*
2. *Calculate its surface area, showing your working.*
3. *Describe and count every plane of symmetry it has.*

4. *Calculate the surface area of a 10 cm cube and compare it with your design.*
5. Write two sentences on which shape uses least material, and why.